Dissertation Title:
STEER FEEL MODELING OF STEER BY WIRE SYSTEM AND ITS SAFETY FACTORS AND IMPLICATIONS

Course No: S2-23_AELZG628T

Course Title: DISSERTATION

Dissertation / Project /Project Work Done by:

Student Name: Aditya Gopal R

BITS ID: 2022HT65151

Degree Program: MTech Automotive Electronics

Research Area: EMBEDDED SYSTEMS DESIGN

Dissertation Work carried out at:

Bosch Global Software Technologies, Coimbatore, Tamil Nadu



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
VIDYA VIHAR, PILANI, RAJASTHAN - 333031.

January 2024

Contents

1. Broad Area of Work 3
2. Background 3
3. Objectives 3
4. Scope of Work 4
5. Plan of Work 4
6. Literature References 5
7. Particulars of the Supervisor and Examiner 5
8. Remarks of the Supervisor 6

1. Broad Area of Work

X-by-wire technologies is a well innovative research area in the field of automation. This research project is intended to study the modelling and simulation of the Steer Feel module of the SbW system and the broad area comprises of the following

- Modelling and simulation: MATLAB to be used to Model and simulate the feedback and control of the Steer Feel
- Risk Assessment: Assessment of risk factor and redundant setup.

2. Background

X-by Wire is used in Active Safety application, such as Brake-By- Wire and Steer-By Wire and also used in certain applications in the power train such as Throttle-by-Wire. Though the system is much dependable, the haptic feedback from the mechanical counterparts is required to give us a much better user comfort and has a significant safety factor included in it. The dissertation is focused mainly on the Steer Feel Module based modelling and simulation.

Modelling, simulation, ISO 26262 Standards, Hazard analysis and Risk Assessment (HARA), Failure mode and Effect Analysis (FMEA) will be used and discussed with respect to SbW in this Dissertation.

3. Objectives

The objectives of my project are as follows:

- Modelling of the Steer Feel required as haptic feedback to the driver.
- Safety Factors such as requirement of a redundant mechanical part and other mechanisms required for SbW health.
- Understand the Functional Safety (ISO26262) and the system safety.
- Risk Assessment and Management

4. Scope of Work

Scope of this dissertation is to model the Steer Feel function of the SbW and Risk assessment and impact of the Steer Feel on the safety of the whole SbW System.

5. Plan of Work

Phases	**Start Date-End Date**	**Work to be done**
Dissertation Outline	13 Jan 2024 – 20 Jan 2024	Literature Review and prepare Dissertation Outline
Design & Development	21 Jan 2024 – 10 Mar 2024	Design & Development Activity
Evaluation	11 Mar 2024 – 25 Mar 2024	Software Testing, User Evaluation & feedback
Final Development	26 Mar 2024-20 Apr 2024	Fixing of open points and betterment done
Review	21 Apr 2024 – 10 May 2024	Submit Dissertation to Supervisor & Additional Examiner for review and feedback
Submission	11 May 2024-17 May 2024	Final Review and submission of Dissertation

6. Literature References

[1] Actuator fault tolerant control for steer-by-wire systems,Chao HuangORCID Icon,Hailong Huang,Fazel Naghdy,Haiping Du &Dazhong Ma, Pages 3123-3134 (2020)

[2] Chao Huang, Fazel Naghdy, Haiping Du, Hailong Huang,Fault tolerant steer-by-wire systems: An overview,. Annual Reviews in Control, Volume 47. (2019)

[3] A. Altby et al. Design and implementation of a fault-tolerant drive-by-wire system (2014)

[4] Research Article A Hysteresis-Based Steering Feel Model for Steer-by-Wire Systems M. Selçuk Arslan, Department of Mechatronics Engineering, Yildiz Technical University, Istanbul Turkey.(2017)

[5] M. K. Salaani, G. J. Heydinger, and P. A. Grygier, "Experimental steering feel performance measures," SAE Technical Papers, 2004

[6] M. Segawa, "Vehicle stability control strategy for steer by wire system," JSAE Review, vol. 22, no. 4, pp. 383–388.

7. Particulars of the Supervisor and Examiner

	Supervisor	**Additional Examiner**
Name	Raja Mohan R	Rajesh R
Qualification	M.E(Embedded Systems)	M.E(VLSI Design)
Designation	Technical Lead	Group Manager
Employing Organization and Location	Bosch Global Software Technologies, Coimbatore	Bosch Global Software Technologies, Coimbatore
Phone No (with STD Code)	+91 98946 87106	+91 94861 29023
Email Address	benzraj1987@gmail.com	rajesh.ramadurai17@gmail.com

8. Remarks of the Supervisor

The study and project in an innovative field has impressive and it involves diligent work to take up in an innovative place. He demonstrates great theoretical knowledge and the capability to understand the system as a whole. Given his involvement in System level V&V , MATLAB Modelling and the Sbw background, with proper guidance ethe task of the dissertation will be achievable.

Information about the Supervisor: Automotive software enthusiast with over 12 years of professional experience. SW Integrations, Verification and Validation(V&V) teams. Software Test Manager for Steer-by-Wire system with experience in System Engineering as Function Owner for ADAS Front Camera system.

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2023-2024

AELZG628T: AUTOMOTIVE ELECTRONICS

STUDENT ID No.	2022HT65151
NAME OF THE STUDENT	Aditya Gopal R
STUDENT'S EMAIL ADDRESS	2022ht65151@wilp.bits-pilani.ac.in
STUDENT'S EMPLOYING ORGANIZATION & LOCATION	Bosch Global Software Technologies, Coimbatore
SUPERVISOR'S NAME	Raja Mohan R
SUPERVISOR'S EMPLOYING ORGANIZATION & LOCATION	Bosch Global Software Technologies, Coimbatore
SUPERVISOR'S EMAIL ADDRESS	benzraj1987@gmail.com
ADDITIONAL EXAMINAER'S NAME	Rajesh R
ADDITIONAL EXAMINER'S EMPLOYING ORGANIZATION & LOCATION	Bosch Global Software Technologies, Coimbatore
ADDITIONAL EXAMINER'S EMAIL ADDRESS	rajesh.ramadurai17@gmail.com
DISSERTATION / PROJECT / PROJECT WORK TITLE	STEER FEEL MODELING OF STEER BY WIRE SYSTEM AND ITS SAFETY FACTORS AND IMPLICATIONS

Please prepare the outline as a separate document with the following sections along with the above identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
2. Background (Relevance of the Project to the current work environment in the employing organization)
3. Objectives
4. Scope of Work (To be done by the student independently)
5. Plan of Work (Work to be done during the semester)
6. Literature References
7. Particulars of the Supervisor and Additional Examiner
8. Remarks of the Supervisor

R. Adity		R. Rajesh
Signature of Student	Signature of Supervisor	Signature of Additional Examiner
Name: Aditya Gopal R	Name: Raja Mohan R	Name: Rajesh R